AP CALCULUS AB

Unit 1 • Lesson 1-3 • Guided notes

Lesson 1-3 • Limits to Infinity and Horizontal Asymptotes

Guided notes • Fill in the blanks and work the examples in the boxes

Learning goals

- Evaluate limits as $x \to \pm\infty$ for rational, radical and exponential expressions.
- Connect limits at infinity to horizontal asymptotes and infinite limits to vertical asymptotes.
- Compare growth rates of logarithmic, power and exponential functions.

Vocabulary. limit at infinity, horizontal asymptote, infinite limit, vertical asymptote, dominant term.

1. Horizontal and vertical asymptotes

Key idea

If $\lim_{x\to\infty} f(x) = L$ or $\lim_{x\to-\infty} f(x) = L$, the line _____ is a horizontal asymptote.
If $\lim_{x\to c^-} f(x)$ or $\lim_{x\to c^+} f(x)$ is $\pm\infty$, the line _____ is a vertical asymptote.

2. Rational functions

Key idea

Compare the degrees of numerator and denominator: smaller on top ⇒ limit ____; equal ⇒ ratio of _____; larger on top ⇒ _____ (no horizontal asymptote).

Example 1

$$\lim_{x\to\infty} \frac{4x^3 - x}{2x^3 + 7x^2}, \quad \lim_{x\to\infty} \frac{5x + 1}{x^2 - 3}, \quad \lim_{x\to-\infty} \frac{x^3}{x^2 + 1}$$

3. Radicals: watch the sign

$\sqrt{x^2} = |x|$, which equals _____ when $x < 0$.

Example 2

Find both horizontal asymptotes of $f(x) = \dfrac{3x - 1}{\sqrt{4x^2 + 5}}$.

4. Growth rates

Key idea

As $x \to \infty$: $\ln x$ grows _____ than x^p $(p > 0)$, which grows _____ than a^x $(a > 1)$. The fastest-growing term controls the limit.

Example 3

$\lim_{x\to\infty} \frac{x^3}{e^x}$, $\lim_{x\to\infty} \frac{3e^x + x^2}{e^x - 1}$, $\lim_{x\to-\infty} e^x$, $\lim_{x\to0^+} \ln x$

Example 4 • Vertical asymptotes

For $f(x) = \frac{x+2}{x^2-4x}$, find the vertical asymptotes and the one-sided limits.

Common mistakes

- Forgetting that $\sqrt{x^2} = -x$ for $x < 0$.
- Thinking $\lim_{x\to-\infty} e^x = -\infty$; it is 0.

Check your understanding

1. $\lim_{x\to\infty} \frac{2-5x^2}{x^2+x}$ = _____. 2. $\lim_{x\to\infty} \frac{x^{100}}{1.01^x}$ = ____. 3. $\lim_{x\to-\infty} \frac{2x}{\sqrt{x^2+5}}$ = _____.